Kitaev Chain Charge Difference

Function reference

Notebook: `kitaev_charge_difference.ipynb`

Repository: `MSH_Tensor_Network_Methods`

Julia 1.13, TensorBinding.jl, ITensors.jl / ITensorMPS.jl

2026-10-29

This document explains every function and helper defined in the notebook: what it computes, the method, inputs and outputs, the numerical settings that matter, and where it is used. All statements are taken from the notebook's code, comments, markdown and saved outputs. Where a library routine of TensorBinding.jl is relevant, its behaviour is quoted from the package source and marked as such.

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1 Overview

1.1 The model as built in the notebook

The notebook studies a spinless one-dimensional Kitaev chain with nearest-neighbour hopping, a site-dependent on-site potential $V(x)$ (called the chemical potential in the notebook) and nearest-neighbour p-wave pairing. Every Hamiltonian is created with three TensorBinding calls:

```
H = TensorBinding.get_Hamiltonian("chain_1d", 1.0; L = Lℓ)   # chain of 2^L sites, hopping 1.0
TensorBinding.add_onsite!(H, Vf)                              # V(x); Vf receives a 0-indexed
 ↪  site
TensorBinding.add_superconductivity!(H, 1.0)                  # pairing amplitude 1.0
```

For spinless fermions `add_superconductivity!` prints *"On-site (s-wave) pairing is forbidden for spinless fermions ... Constructing nearest-neighbour p-wave instead"*, so the pairing is nearest-neighbour p-wave with $\Delta = 1$. The resulting object is a `TBHamiltonian` (printed as `L=6, N=64 +BdG, maxlinkdim=5`) whose field `.mpo` is the Bogoliubov–de Gennes (BdG) Hamiltonian as an MPO, `.sites` its site indices, `.L` the number of position bits and `.N`$= 2^L$ the number of lattice sites.

Sign conventions of the library Hamiltonian. Cell 35 reads the dense BdG matrix of the hopping-plus-pairing part off the MPO (`dense_bdg`, §7) and compares it element by element with

$$H_{\mathrm{BdG}} = \begin{pmatrix} h & \Delta \\ \Delta^\dagger & -h^T \end{pmatrix}, \qquad h_{x,x\pm1} = +t, \quad \Delta_{x,x+1} = +\Delta, \quad \Delta_{x+1,x} = -\Delta, \quad \Delta_{x,x} = 0,$$

with $t = \Delta = 1$ (maximum deviation 3.6×10^{-15}). In second quantization,

$$H = \sum_x \Big[t\,(c_x^\dagger c_{x+1} + \mathrm{h.c.}) + \Delta\,(c_x^\dagger c_{x+1}^\dagger + \mathrm{h.c.}) + V(x)\, c_x^\dagger c_x \Big],$$

with the on-site term entering the BdG matrix as $\tau_z \otimes \mathrm{diag}\, V$. The hopping thus enters with a $+$ sign. The gauge transformation $c_x \to (-1)^x c_x$ maps $t \to -t$ (and $\Delta \to -\Delta$), so the spectrum is that of the usual Kitaev chain with hopping $-t$, topological for $|V| < 2|t|$.

Nambu and position qubits. A BdG single-particle state is an MPS on $L+1$ two-level sites. Site 1 is the Nambu index ($|0\rangle$ = particle, $|1\rangle$ = hole; "Nambu is first index" in the code). Sites $2, \dots, L+1$ are the binary digits $b_1 \dots b_L$ of the position, most significant first, with

$$x = 1 + \sum_{k=1}^{L} b_k\, 2^{L-k}, \qquad x = 1, \dots, N = 2^L.$$

A BdG vector $\psi = (u, v)$ thus has particle amplitudes $u(x) = \langle 0, x|\psi\rangle$ and hole amplitudes $v(x) = \langle 1, x|\psi\rangle$. The on-site potential enters as $\tau_z \otimes \mathrm{diag}\, V(x)$; the notebook checks this numerically ($\langle 0, x|H|0, x\rangle = V(x)$, see `charge_difference` and the time-dependent section).

Particle–hole conjugation. The BdG Hamiltonian anticommutes with $C = \tau_x K$ (K = complex conjugation). On an MPS the notebook implements C as `dag(ψ)` (complex conjugation of all tensors) followed by τ_x applied to the Nambu site. For $\psi = (u, v)$ this gives $C\psi = (v^*, u^*)$. If $H\psi = E\psi$ then $HC\psi = -E\,C\psi$, so states come in $\pm E$ pairs. A Majorana state is self-conjugate, $C\gamma = \gamma$.

Parameters.

Quantity	Static part (§3–6)	Time-dependent part (§7)
Position bits L	`Lℓ = 6`	`Lt = 5`
Sites $N = 2^L$	64	32
Hopping, pairing	1.0, 1.0	1.0, 1.0
Topological potential	`V_in = -1.0`	`μ_topo_t = -0.4`
Trivial potential	`V_out = -4.0`	`μ_triv_t = -5.6`
Topological region	centred, length ℓ (`region(ℓ)`)	whole chain at $t = 0$, site 16 at $t = T$

With hopping 1 the notebook's markdown refers to the topological criterion $|\mu| < 2t$; $V = -1$ and -0.4 are inside it, -4 and -5.6 outside.

1.2 Logical states and the charge difference

Following the notebook's markdown, the two logical states $|0\rangle$ (fermion mode empty) and $|1\rangle$ (mode occupied) are represented by their BdG correlation matrices

$$\Gamma_{ab} = \langle \Psi_b^\dagger \Psi_a \rangle, \qquad \Psi = (c_1, \dots, c_N, c_1^\dagger, \dots, c_N^\dagger).$$

Since H is quadratic, Γ fully characterises the BCS state and is the projector on its occupied BdG states. Both logical states share the filled bulk projector P_{bulk} and differ in which member of the near-zero pair $\pm E$ is filled:

$$\Gamma_0 = P_{\text{bulk}} + |\bar{\psi}_f\rangle\langle\bar{\psi}_f|, \qquad \Gamma_1 = P_{\text{bulk}} + |\psi_f\rangle\langle\psi_f|, \qquad \bar{\psi}_f = C\psi_f.$$

The local density is the particle–particle diagonal, $n(x) = \langle c_x^\dagger c_x \rangle = \langle 0, x|\Gamma|0, x\rangle$. Because $\Gamma_1 - \Gamma_0 = |\psi_f\rangle\langle\psi_f| - |\bar{\psi}_f\rangle\langle\bar{\psi}_f|$ exactly,

$$\delta n(x) = n_1(x) - n_0(x) = |u_f(x)|^2 - |u_{\bar{f}}(x)|^2 = |u_f(x)|^2 - |v_f(x)|^2, \qquad \Delta Q = \sum_x \delta n(x) = \langle \psi_f | \tau_z | \psi_f \rangle. \tag{1}$$

The second equality uses $u_{\bar{f}} = v_f^*$ from $C\psi = (v^*, u^*)$. The physical charge is $-e\,\delta n$; plots label $|\Delta Q|$ in units of $-e$.

1.3 Workflow of the notebook

1. Helpers to build position basis states (§2).
2. "Good" single-particle states: penalized DMRG for the near-zero doublet, localisation of the Majoranas by diagonalising the position operator, self-conjugation, pairing into ψ_f (§3).
3. KPM local density of states and the bulk gap estimate `kpm_gap` (§4).
4. McWeeny-purified bulk projector, Γ_0, Γ_1, consistency checks and local densities (§5).
5. ΔQ versus the length ℓ of the topological region, by two methods (§6).
6. Time-dependent smoothstep ramp that moves the Majoranas together: TDVP evolution of ψ_f, $\Delta Q(t)$, $\delta n(x, t)$, and full densities with a TDVP-evolved or McWeeny (adiabatic) bulk (§7).

2 Position basis states

`to_bits(n, L)`

```
function to_bits(n::Integer, L::Integer)
    return [parse(Int, c) for c in lpad(string(n; base=2), L, '0')]
end
```

Returns the L-digit binary representation of n as a `Vector{Int}`, most significant bit first. `string(n; base=2)` gives the binary string, `lpad` pads it with `'0'` on the left to length L, and each character is parsed to an integer. Callers pass `x - 1` to convert the 1-indexed site x to $0, \ldots, N-1$.

`basis_state(sites, bits)`

```
function basis_state(sites, bits::Vector{Int})
    return MPS([onehot(sites[i] => bits[i] + 1) for i in eachindex(sites)])
end
```

Builds the product MPS $|b_0 b_1 \ldots b_L\rangle$. `onehot(s => 1)` is $[1,0] = |0\rangle$ and `onehot(s => 2)` is $[0,1] = |1\rangle$; the `+1` converts bits to Julia's 1-based indices. The bond dimension is 1. Throughout the notebook it is called as `basis_state(sites, vcat([nb], to_bits(x - 1, L)))`, where the leading bit `nb` selects the Nambu sector ($0 =$ particle, $1 =$ hole). It is the building block of `site_weights`, `pbasis`, `pb`, `pbt`, `hbt` and `bdg_basis`.

3 The fermion mode from the Majorana doublet

The markdown lists the procedure for "good" single-particle states: (1) find orthonormal $\{\psi_1, \psi_2\}$ spanning the near-zero subspace with DMRG; (2) localise the Majorana wavefunctions; (3) enforce self-conjugation $C\gamma = \gamma$; (4) pair the Majoranas into a fermion.

`doublet(Hmpo, sites; η=1e-2, seed=1234, cutoff=1e-20)`

```
K = TensorBinding.build_K(Hmpo, sites, 0.0, η; cutoff_K=cutoff)
sw = (nsweeps=15, maxdim=[10, 20, 50, 100, 150], cutoff=cutoff,
      noise=[1e-6, 1e-7, 1e-8, 0.0], outputlevel=0)
Random.seed!(seed)
_, ψ1 = dmrg(K, random_mps(sites; linkdims=10); sw...)
_, ψ2 = dmrg(K, [ψ1], random_mps(sites; linkdims=10); weight=10.0, sw...)
ψ1 /= norm(ψ1)
ψ2 = add2(ψ2, -inner(ψ1, ψ2) * ψ1); ψ2 /= norm(ψ2)
```

Computes two orthonormal BdG states spanning the near-zero doublet of H.

Method. The states with smallest $|E|$ are not the ground state of H, so DMRG is run on the positive operator

$$K = (H - \omega)^2 + \eta^2, \qquad \omega = 0,$$

built by `TensorBinding.build_K` (from the package source: $H - \omega\mathbb{1}$, squared with `apply` using `cutoff_K` and the default `maxdim_K = 200`, then $+\eta^2\mathbb{1}$). Eigenvectors of K are those of H, with eigenvalue $E^2 + \eta^2$; the lowest two are the $\pm E$ doublet.

Role of η. `build_K` serves TensorBinding's spectral DMRG, where $K(\omega, \eta)$ is minimised at a chosen energy ω and η is a broadening: $\min\langle K\rangle = \eta^2$ when ω hits an eigenvalue of H. Here only $\omega = 0$ is used, and η^2 shifts every eigenvalue of K by the same constant, so it changes neither the states DMRG finds nor its convergence (Krylov iterations are invariant under a constant shift). The notebook's comment records a check on the $\ell = 6$ chain: with $\eta = 0$, E_{mode} and E_{bulk} agree to all printed digits and the overlap of the two fermion modes is 1. `lowest_bulk` subtracts η^2 again. The value

$\eta = 10^{-2}$ is kept for consistency with the MSH notebooks. `cutoff_K` must stay small (here equal to `cutoff`): the package default 10^{-8} can truncate the η^2 term of the MPO.

DMRG runs. The first DMRG gives ψ_1. The second DMRG adds a penalty of weight 10 against ψ_1 (ITensor's `dmrg(H, [ψ1], ψ0; weight)`), so it converges to the next eigenvector of K. A final Gram–Schmidt step removes any residual overlap with ψ_1.

Settings. 15 sweeps; bond dimension ramped through 10, 20, 50, 100, 150; noise $10^{-6} \to 10^{-8} \to 0$; truncation cutoff 10^{-20}; random initial MPS of bond dimension 10 with a fixed seed (1234) for reproducibility.

Returns `(ψ1, ψ2)`. **Used by** `fermion_mode`, `initial_mode`, `final_branches`.

`lowest_bulk(Hmpo, sites, ψ1, ψ2; η=1e-2, seed=1234, cutoff=1e-20)`

```
_, φ = dmrg(K, [ψ1, ψ2], random_mps(sites; linkdims=10); weight=10.0, sw...)
φ = add2(φ, -inner(ψ1, φ) * ψ1); φ = add2(φ, -inner(ψ2, φ) * ψ2); φ /= norm(φ)
return φ, sqrt(max(real(inner(φ', K, φ)) - η^2, 0.0))
```

Computes the lowest bulk state, i.e. the eigenstate with the next-smallest $|E|$ after the doublet (the bulk band edge), and its energy. It runs a third DMRG on the same K and with the same sweep settings as `doublet`, penalised against both ψ_1 and ψ_2, then orthogonalises against them. The energy follows from $\langle K \rangle$:

$$|E_{\text{bulk}}| = \sqrt{\max(\langle\varphi|K|\varphi\rangle - \eta^2,\, 0)}.$$

Penalty weight. The code comment states that the weight (10) must exceed $E_{\text{bulk}}^2 + \eta^2$, which holds for $|E_{\text{bulk}}| < 3$; otherwise the penalised doublet states would not lie above the bulk state in the penalised problem.

Returns `(φ, |E_bulk|)`. **Used by** `fermion_mode` when `gap_factor` is given.

`add2(a, b)`

```
add2(a, b) = +(a, b; cutoff=1e-16)
```

MPS addition with truncation cutoff 10^{-16}. All linear combinations of MPS in the notebook use it, so that sums are compressed without losing accuracy.

`fermion_mode(Hmpo, sites, L; η=1e-2, seed=1234, cutoff=1e-20, gap_factor=nothing)`

Computes the BdG wavefunction of the ordinary fermion $\psi = (\gamma_L + i\gamma_R)/\sqrt{2}$ built from the two Majoranas, its energy, the Majoranas themselves, and optionally the bulk edge.

Step 1: doublet. `ψ1, ψ2 = doublet(...)`.

Step 2: localisation. DMRG returns an arbitrary basis of the doublet. The Majoranas are the combinations that diagonalise the position operator in that subspace. With the bit projector $\hat{n}_k = |1\rangle\langle 1|$ on position bit k (MPS site $k+1$),

$$\hat{x} = \mathbb{1} + \sum_{k=1}^{L} 2^{L-k}\, \hat{n}_k, \qquad S_{ab} = \langle\psi_a|\psi_b\rangle, \quad X_{ab} = \langle\psi_a|\hat{x}|\psi_b\rangle, \quad a,b \in \{1,2\}.$$

```
nk(k) = ITensor([0.0 0.0; 0.0 1.0], sites[k+1]', sites[k+1])
on_site(op, ψ, j) = (φ = copy(ψ); φ[j] = noprime(op * φ[j]); φ)
S = [inner(a, b) for a in ψs, b in ψs]
X = S + sum(2.0^(L - k) * [inner(a, on_site(nk(k), b, k + 1)) for a in ψs, b in ψs] for k in
 ↪  1:L)
F = eigen(Hermitian((X + X') / 2), Hermitian((S + S')/2))
```

The S term supplies the $\mathbb{1}$ in $\hat{x}$. The generalized Hermitian eigenproblem $Xc = \lambda Sc$ is solved (symmetrised to remove round-off). Eigenvalues are ascending, so column 1 (smaller $\langle x \rangle$) gives γ_L and column 2 gives γ_R.

Step 3: self-conjugation. Each combination carries an arbitrary phase: $C\gamma = e^{i\theta}\gamma$.

```
τx_s = ITensor([0.0 1.0; 1.0 0.0], sites[1]', sites[1])   # Nambu is first index
C(ψ) = (φ = dag(ψ); φ[1] = noprime(τx_s * φ[1]); φ)
fix(γ) = exp(im * angle(inner(γ, C(γ))) / 2) * γ
```

With $\theta = \arg\langle\gamma|C\gamma\rangle$, the state $\gamma' = e^{i\theta/2}\gamma$ satisfies $C\gamma' = e^{-i\theta/2}e^{i\theta}\gamma = \gamma'$.

Step 4: pairing and branch choice.

$$\psi = \frac{\gamma_L + i\gamma_R}{\|\gamma_L + i\gamma_R\|}, \qquad E = \langle\psi|H|\psi\rangle, \qquad \text{if } E < 0 : \ \psi \leftarrow C\psi,\ E \leftarrow -E.$$

The comment notes that $\|\gamma_L + i\gamma_R\|^2 = 2$, so normalising before computing E avoids getting $2E$. By convention $|1\rangle$ is the $+E$ state, so the returned ψ is always the positive-energy member; $C\psi$ is its $-E$ partner.

Step 5: gap check (optional). If `gap_factor` is set, `lowest_bulk` gives $|E_{\text{bulk}}|$, and a warning is issued if $|E_{\text{bulk}}| <$ gap_factor $\cdot\, E$, i.e. if the doublet does not sit well below the bulk. Otherwise `E_bulk = NaN`.

Returns `(ψ, E, γL, γR, E_bulk)`. **Used** in cell 9 for the $\ell = 6$ chain `H0` and inside `charge_difference` and `charge_difference_method_2`. The time-dependent part uses a copy without the energy and gap steps, `initial_mode`.

3.1 Static setup and profile helpers (cells 8–12)

Cell 8 defines `Lℓ = 6`, `Nℓ = 64`, `V_in, V_out = -1.0, -4.0` and

```
region(ℓ) = ((Nℓ - ℓ) ÷ 2 + 1):((Nℓ - ℓ) ÷ 2 + ℓ)
Vf(n) = (n + 1) in reg ? V_in : V_out
```

`region(ℓ)` is the centred range of ℓ 1-indexed sites. `Vf` receives the 0-indexed site from `add_onsite!`, hence `n + 1`. The notebook sets `ℓ = 6`, so `reg = 30:35`: in `H0` the six sites $x = 30 \ldots 35$ are topological, with the Majoranas at the two ends of that region. Cell 9 calls `fermion_mode(H0.mpo, H0.sites, Lℓ)`.

`site_weights(γ), C(ψ), τx_s`

```
site_weights(γ) = [sum(abs2(inner(basis_state(H0.sites, vcat([nb], to_bits(x - 1, H0.L))), γ))
 ↪  for nb in 0:1)
                   for x in 1:H0.N]
```

`site_weights` returns $w(x) = |u(x)|^2 + |v(x)|^2$ for $x = 1, \ldots, N$: the probability density of a BdG state summed over the Nambu index. It gives the profiles of γ_L, γ_R, ψ and $\bar{\psi} = C\psi$ plotted in Figure 1. `τx_s` and `C` in cell 10 are global copies of the particle–hole operator defined inside `fermion_mode`.

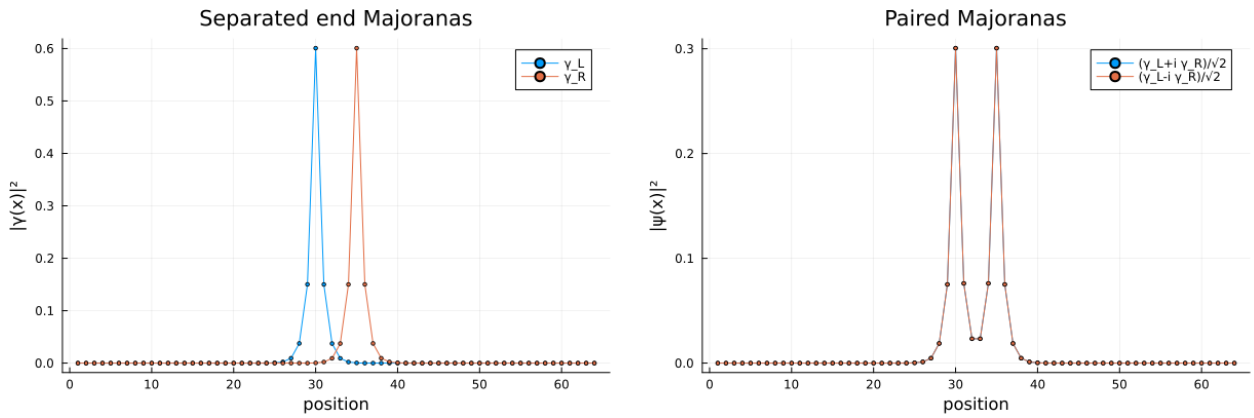


Figure 1: Left: $|\gamma_L(x)|^2$ and $|\gamma_R(x)|^2$ from `site_weights`. Right: $|\psi(x)|^2$ for $(\gamma_L \pm i\gamma_R)/\sqrt{2}$, i.e. ψ_f and $C\psi_f$, which have identical profiles. Both panels are for the notebook's `H0` with $\ell = 6$ (sites 30–35 topological): γ_L peaks at $x = 30$ and γ_R at $x = 35$, the two ends of the topological region, and ψ_f has weight at both.

4 KPM local density of states and the gap estimate

LDOS call (cell 13).

```
Ncheb = 200; mdim = 60
omega = range(-4.0, 4.0; length=1500)
ldos = TensorBinding.get_ldos_spatial(H0, Ncheb, collect(omega); num_x=H0.N, mode=:mpo,
 ↪  kernel=:jackson,
                                      maxdim=mdim, verbose=false)
```

The library routine computes the spatially resolved LDOS $A(x, \omega)$ with the kernel polynomial method (KPM): 200 Chebyshev moments, MPO mode, bond dimension 60, on 1500 energies in $[-4, 4]$ for all 64 sites. The printed log shows that the spectral bounds were estimated by DMRG ($E_{\min} = -5.990$, $E_{\max} = 5.990$, scale 6.589); the Hamiltonian is rescaled to $[-1, 1]$ with these bounds. The Chebyshev moments are damped with the Jackson kernel (the library default, written out in the call; Lorentz, Fejér and Dirichlet kernels are the alternatives); see Weiße *et al.*, Rev. Mod. Phys. **78**, 275 (2006). Because `H0` has a Nambu index, the library switches the Nambu projection on automatically (printed info line); with the sector selector `proj_nambu` left at `nothing` it sums the particle and hole sectors, so $A(x, \omega) = \sum_n (|u_n(x)|^2 + |v_n(x)|^2)\, \delta_{\mathrm{KPM}}(\omega - E_n)$, symmetric in ω. The result is a matrix with one row per energy and one column per site; it is plotted in Figure 2.

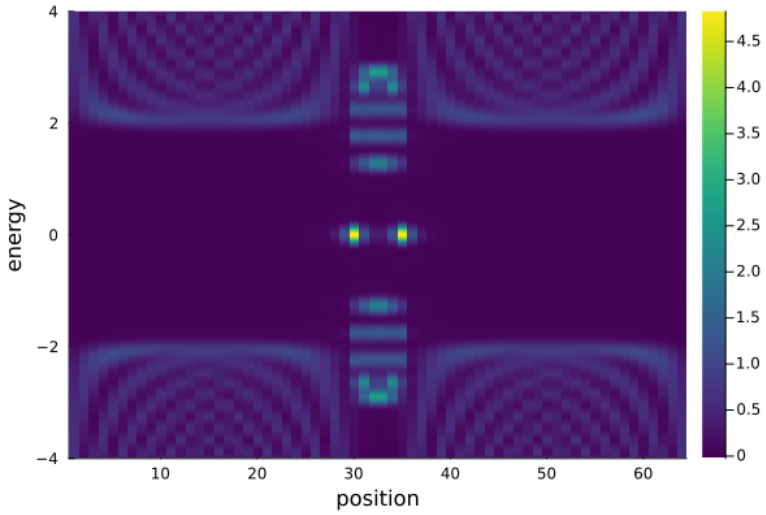


Figure 2: KPM LDOS $A(x, \omega)$ of `H0` ($\ell = 6$). The near-zero pair ($E_{\text{mode}} = 0.019$) appears at $x = 30$ and $x = 35$, the ends of the topological region; states of the region itself lie at $|\omega| \gtrsim 1.2$, and the bulk bands of the trivial chain start near $|\omega| \approx 2$.

`kpm_gap(ldos, omega, E_mode; mode_width=0.3)`

```
ω = collect(omega)
dos = vec(sum(ldos; dims=2))
D(w) = (dos[argmin(abs.(ω .- w))] + dos[argmin(abs.(ω .+ w))]) / 2
ws = ω[ω .> abs(E_mode) + mode_width]
Ds = D.(ws)
noise = 0.1 * maximum(Ds)
j = findfirst(k -> Ds[k] > noise && Ds[k] >= Ds[k+1], 1:length(Ds)-1)     # first bulk peak
half = Ds[j] / 2
i = findlast(k -> Ds[k] < half, 1:j)                                       # rising flank
return ws[i] + (half - Ds[i]) * (ws[i+1] - ws[i]) / (Ds[i+1] - Ds[i])
```

Computes the bulk gap edge E_{gap} from the KPM LDOS.

Method.

1. Total DOS: $\text{DOS}(\omega) = \sum_x A(x, \omega)$, symmetrised as $D(w) = [\text{DOS}(w) + \text{DOS}(-w)]/2$ (BdG symmetry), evaluated on the nearest grid points.
2. Only energies $w > |E_{\text{mode}}| +$ mode_width are kept, so the mode's own peak is not mistaken for the band edge. The comment sets `mode_width` $\approx 3\times$ the KPM resolution $\pi \cdot \text{scale}/N_{\text{cheb}} \approx 0.1$.
3. The first bulk peak is the first local maximum ($D_k \geq D_{k+1}$) above a noise floor of 10% of the maximum.
4. E_{gap} is the half-maximum point on the rising flank of that peak, found by linear interpolation between the last grid point below half maximum and the next one.

The half-maximum is used because, per the comment, the KPM-broadened peak sits about 0.1 above the true edge.

Returns E_{gap} (a `Float64`). **Used** in cell 16 to set the Fermi level. For `H0` it gives $E_{\text{gap}} = 1.154$; the DMRG value from `lowest_bulk` for the same chain ($\ell = 6$ in the sweep of §6) is 1.275.

5 Bulk projector and logical states

Cell 16 builds P_{bulk}, Γ_0 and Γ_1 as MPOs and checks them. It defines no named function apart from two one-line helpers (`Cop`, `pbasis`); the steps are described in order.

Fermi level. P_{bulk} must contain all negative-energy bulk states and neither member of the $\pm E$ mode. The Fermi level is therefore placed midway in the window between the bulk edge and the lower member of the mode:

$$\epsilon_F = -\frac{E_{\text{mode}} + E_{\text{gap}}}{2},$$

with E_{mode} the energy returned by `fermion_mode` and E_{gap} from `kpm_gap`. Printed: $E_{\text{mode}} = 0.018763$, $E_{\text{gap}} = 1.15365$, $\epsilon_F = -0.58621$.

McWeeny purification.

```
ρ_bulk = TensorBinding.mcweeny_purify(H0; ϵF=ϵF, maxiters=60, maxdim=200, cutoff=1e-20,
 ↪  tol=1e-8)
```

McWeeny purification iterates $\rho \to 3\rho^2 - 2\rho^3$, which sends eigenvalues above $\frac{1}{2}$ to 1 and those below to 0 (the comment in `mcweeny_bulk`, §7). Starting from a linear guess in H whose eigenvalues exceed $\frac{1}{2}$ exactly for $E < \epsilon_F$, the result is the projector $P_{\text{bulk}} = \sum_{\text{bulk } E_n<0} |\psi_n\rangle\langle\psi_n|$. From the package source, the `TBHamiltonian` method uses the initial guess $\rho_0 = (\frac{1}{2} + \frac{\epsilon_F + \text{center}}{2\,\text{scale}})\mathbb{1} - \frac{H}{2\,\text{scale}}$ with the Hamiltonian's stored KPM `center` and `scale`.

Accuracy settings. The markdown states that accuracy needs `cutoff=1e-20` and `tol=1e-8`. With looser settings, or by iterating past convergence, P_{bulk} drifts to an error of about 10^{-5}, because McWeeny does not correct errors that mix occupied and empty states, so they accumulate.

Logical states.

```
τx = ITensor([0.0 1.0; 1.0 0.0], H0.sites[1]', H0.sites[1])
Cop(ψ) = (φ = dag(ψ); φ[1] = noprime(τx * φ[1]); φ)
psi_bar = Cop(psi)
Γ0 = +(ρ_bulk, outer(psi_bar', psi_bar); cutoff=1e-20)
Γ1 = +(ρ_bulk, outer(psi', psi); cutoff=1e-20)
```

`Cop` is again $C = \tau_x K$. `outer(ψ', ψ)` is the rank-one MPO $|\psi\rangle\langle\psi|$.

Checks (printed).

- $\operatorname{Tr} P_{\text{bulk}} = 62.999999999993$, equal to the rank $N - 1 = 63$ expected for the projector.
- Mode occupation $\langle\psi|\Gamma|\psi\rangle$: 9.0×10^{-13} for $|0\rangle$, 1.0000000000009 for $|1\rangle$. $\langle\bar{\psi}|\Gamma|\bar{\psi}\rangle$: 0.99999999999997 for $|0\rangle$, -1.7×10^{-14} for $|1\rangle$.
- Many-body energies $E_{0,1} = \frac{1}{2}\operatorname{Tr}(H\Gamma_{0,1})$ (up to a constant), evaluated exactly as `inner(H0.mpo, Γ)/2` since `inner(A,B)` $= \operatorname{Tr}(A^\dagger B)$: $E_1 - E_0 = 0.0187632080507$, compared with $E = 0.0187632080507$ from `fermion_mode`. The comment notes that this difference is small only when the MZMs are far apart; here $\ell = 6$.

`pbasis(x)`

```
pbasis(x) = basis_state(H0.sites, vcat([0], to_bits(x - 1, H0.L)))
n0 = [real(inner(pbasis(x)', Γ0, pbasis(x))) for x in 1:H0.N]
```

The particle-sector basis state at site x (the BdG vector of c_x). It is used to read the densities $n_{0,1}(x) = \langle 0, x|\Gamma_{0,1}|0, x\rangle$ shown in Figure 3. A code comment notes that the difference δn should be computed directly from ψ_f, Eq. (1), rather than by subtracting two purified MPOs.

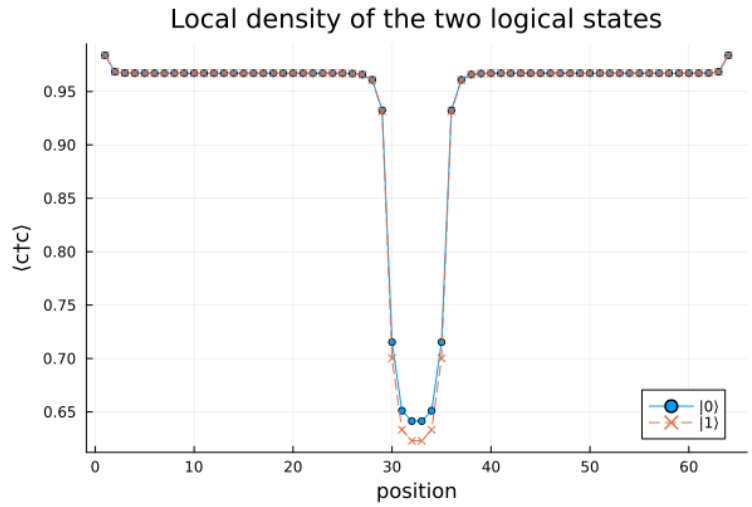


Figure 3: Local densities $n_0(x)$ and $n_1(x)$ of the two logical states of `H0` ($\ell = 6$). They differ only in the topological region $x = 30 \ldots 35$; the total difference is $\Delta Q = -0.105$ (the $\ell = 6$ entry of §6).

6 Charge difference versus length of the topological region

One topological region ($V = -1$) of length ℓ is centred in the 64-site chain, with $V = -4$ elsewhere. The MZMs sit at the two ends of the region, so shrinking ℓ pushes them together. For each ℓ the code comment lists the pipeline: zero-mode pair from penalized DMRG on $K \to$ gap check (third DMRG) $\to$ localise with $\hat{X} \to$ self-conjugate $\gamma \to$ pair into $\psi_f \to$ keep the $+E$ member $\to \Delta Q = \langle \psi_f | \tau_z | \psi_f \rangle$. Cell 18 repeats the definitions of `Lℓ`, `Nℓ`, `V_in`, `V_out` and `region`.

`τz_on(ψ, s), ΔQ_mode(ψ, s)`

```
τz_on(ψ, s) = (φ = copy(ψ); φ[1] = noprime(ITensor([1.0 0.0; 0.0 -1.0], s[1]', s[1]) * φ[1]);
 ↪  φ)
ΔQ_mode(ψ, s) = real(inner(ψ, τz_on(ψ, s)))
```

`τz_on` applies τ_z to the Nambu site. `ΔQ_mode` returns $\langle \psi | \tau_z | \psi \rangle = \sum_x (|u(x)|^2 - |v(x)|^2)$, which for $\psi = \psi_f$ is the charge difference ΔQ of Eq. (1). The site list `s` only supplies the Nambu index. Since C maps $(u, v) \to (v^*, u^*)$, $\Delta Q(C\psi) = -\Delta Q(\psi)$. Used by `charge_difference` and throughout the time-dependent section.

`charge_difference(ℓ; η=1e-2, seed=1234, cutoff=1e-20, gap_factor=2.0)`

```
reg = region(ℓ)
Vf(n) = (n + 1) in reg ? V_in : V_out
H = TensorBinding.get_Hamiltonian("chain_1d", 1.0; L = Lℓ)
TensorBinding.add_onsite!(H, Vf); TensorBinding.add_superconductivity!(H, 1.0)
pb(x) = basis_state(s, vcat([0], to_bits(x - 1, Lℓ)))
onsite_err = maximum(abs(real(inner(pb(x)', H.mpo, pb(x))) - Vf(x - 1)) for x in 1:Nℓ)
onsite_err > 1e-6 && @warn "ℓ = $ℓ: on-site potential of H deviates from Vf by $onsite_err"
psi, E, γL, γR, E_bulk = fermion_mode(H.mpo, s, Lℓ; η=η, seed=seed, cutoff=cutoff,
 ↪  gap_factor=gap_factor)
return (ΔQ = ΔQ_mode(psi, s), E = E, E_bulk = E_bulk, psi = psi, s = s)
```

Method 1. Builds the Hamiltonian for region length ℓ, checks that the diagonal particle element of the MPO reproduces $V(x)$ at every site to 10^{-6} (the MPO of the on-site term is compressed by the library), calls `fermion_mode` with the gap check enabled (`gap_factor = 2`: warn if $|E_{\text{bulk}}| < 2E$), and

returns a named tuple $(\Delta Q, E, E_{\text{bulk}}, \psi_f, s)$ with ΔQ for the $+E$ member ($|1\rangle$). The mode `psi` and its site list `s` are kept so that `charge_difference_method_2` can reuse them.

Used in cell 20 for $\ell = 1, \ldots, 40$; the cell prints the table $(\ell, \Delta Q, E, |E_{\text{bulk}}|)$ of §8. Apart from that table the output contains only the library's p-wave info lines, no on-site or gap-check warnings.

`charge_difference_method_2(r)`

```
psi, s = r.psi, r.s
psibar = C(psi)
upart(ψ) = [abs2(inner(pb(x),ψ)) for x in 1:Nℓ]
n_density = upart(psi) - upart(psibar)
charge_diff = sum(n_density)
```

Method 2. Takes the entry `r` of the ℓ sweep (`sweep_ℓ`), i.e. the same DMRG mode ψ_f as method 1, so no second DMRG is run, and evaluates ΔQ as the sum of the density difference read off the projectors, $\delta n(x) = |u_f(x)|^2 - |u_{\bar{f}}(x)|^2$ with $\bar{\psi}_f = C\psi_f$ (the comment calls this "using projectors Γ_0 and Γ_1"; the bulk cancels, so only ψ_f and $C\psi_f$ enter). By Eq. (1) it equals method 1 analytically: $u_{\bar{f}} = v_f^*$, so $\sum_x (|u_f|^2 - |u_{\bar{f}}|^2) = \langle \psi_f | \tau_z | \psi_f \rangle$. Method 2 is therefore an independent evaluation of the same number. It returns a single number. Used in cell 24 for every entry of the sweep; the cell prints $\max_\ell |\Delta Q_\ell^{(1)} - \Delta Q_\ell^{(2)}| = 5.6 \times 10^{-16}$, and the two curves coincide in Figure 4.

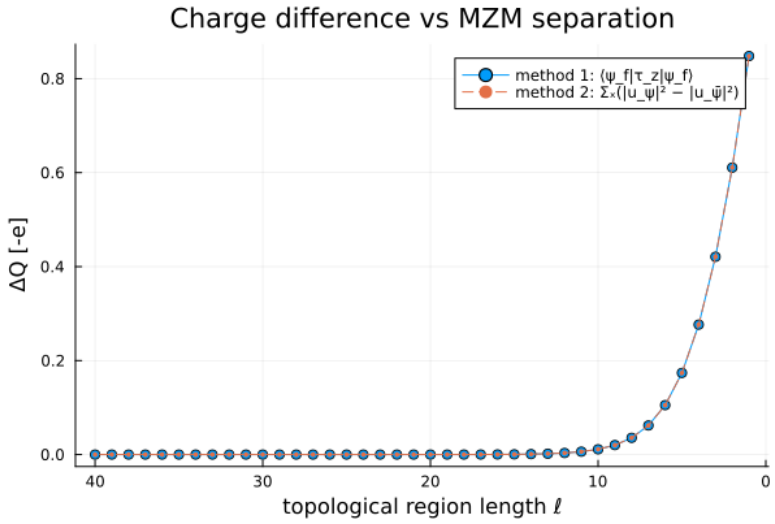


Figure 4: $|\Delta Q|$ (units of $-e$) versus the topological region length ℓ, axis reversed so that the MZMs approach each other from left to right. Method 1 (`charge_difference`) and method 2 (`charge_difference_method_2`, dashed) are plotted on top of each other and agree to 6×10^{-16}.

7 Time-dependent ramp

7.1 Protocol

From the markdown: at $t = 0$ the 32-site chain ($L_t = 5$) is topological ($\mu_{\text{topo}} = -0.4$) with MZMs at its ends. Sites are ramped to $\mu_{\text{triv}} = -5.6$ from the chain ends inward, each with a small delay, which carries the MZMs to the middle. At $t = T$ only the centre site ($\ell_{\text{center}} = 1$, $x = 16$) is topological, so the two MZMs overlap and the logical states differ in charge.

$$\mu(x,t) = \mu_{\text{topo}} + (\mu_{\text{triv}} - \mu_{\text{topo}})\, s(q_i(t)), \qquad q_i(t) = \frac{t}{T}[1 + \alpha(N_{\text{leg}} - 1)] - \alpha i, \qquad s(\tau) = 3\tau^2 - 2\tau^3,$$

with i the distance of x from the nearest chain end, $\alpha = 0.025$ and s clamped to $[0,1]$. The $N_t - \ell_{\text{center}} = 31$ ramped sites cannot split evenly: the left leg has 15 sites ($x = 1 \dots 15$), the right leg 16 ($x = 17 \dots 32$), and $N_{\text{leg}} = 16$ is the longer leg.

- Site-delayed ramp: site i starts at $t = \alpha i T/[1 + \alpha(N_{\text{leg}} - 1)]$ and ramps over the duration $T/[1 + \alpha(N_{\text{leg}} - 1)]$; the trivial region grows inward from both ends at the same rate. The innermost site of the right leg ($i = 15$) finishes at $t = T$, that of the left leg ($i = 14$) at $t/T = (1+14\alpha)/(1+15\alpha) \approx 0.98$.
- A uniform ramp would cross $|\mu| = 2t$ everywhere at once and close the bulk gap. With the delay the instantaneous bulk gap stays ≥ 0.83 (minimum near $t/T \approx 0.43$).
- The smoothstep starts and stops with zero velocity.

What is evolved. $\Gamma_1(t) - \Gamma_0(t) = |\psi_f\rangle\langle\psi_f| - |C\psi_f\rangle\langle C\psi_f|$ at all times, so $\Delta Q(t) = \langle\psi_f(t)|\tau_z|\psi_f(t)\rangle$ needs only the single BdG vector ψ_f, evolved with $i\partial_t\psi = H(t)\psi$ by TDVP. At $t = T$ the weight of ψ_f outside the instantaneous $\pm E$ doublet is the leakage into bulk quasiparticles (the non-adiabatic error). Full densities add the shared bulk, either TDVP-evolved (all $N_t - 1$ filled states) or the adiabatic McWeeny projector of $H(t)$.

7.2 Initial mode and final branches (cell 27)

`C_t(ψ)`

```
C_t(ψ) = (φ = dag(ψ); s = siteind(φ, 1); φ[1] = noprime(ITensor([0.0 1.0; 1.0 0.0], s', s) *
 ↪  φ[1]); φ)
```

Particle–hole conjugation $C = \tau_x K$ that reads the Nambu index from the MPS's own first site, so it does not depend on a global site list. The comment states that the functions of this cell work for any chain length and read no globals.

`initial_mode(Hmpo, sites, L)`

Same as steps 1–4 of `fermion_mode`: `doublet`, localisation with $\hat{X}$, `fix` for $C\gamma = \gamma$ (using `C_t`), and $\psi = (\gamma_L + i\gamma_R)/\|\cdot\|$. Differences: it uses the ordinary eigenproblem `eigen(Hermitian((X + X') / 2))` (the doublet is already orthonormal, so $S = \mathbb{1}$ up to round-off), it does not compute E and does not choose the $+E$ branch, and it uses the default η, seed and cutoff of `doublet`. Returns the normalised ψ.

Used once (cell 29): `ψf_init = initial_mode(H_of_t_alt(0.0, 1.0, smoothstep), st, Lt)`; $H(0)$ is the same for every T and ramp. Printed checks: $\|H\psi\| = 1.12 \times 10^{-12}$ (the MZMs at the ends of the 32-site topological chain are at zero energy to this precision) and $\Delta Q(0) = 4.6 \times 10^{-15}$.

`final_branches(Hmpo, sites)`

```
ψ1, ψ2 = doublet(Hmpo, sites)
Hm = [inner(a', Hmpo, b) for a in ψs, b in ψs]
F = eigen(Hermitian((Hm + Hm') / 2))
mk(c) = add2(c[1] * ψ1, c[2] * ψ2)
return mk(F.vectors[:, 2]), mk(F.vectors[:, 1]), F.values[2]    # ψ₊, ψ₋, E₊
```

Instantaneous $\pm E$ eigenstates at the end of the ramp. The 2×2 matrix of H in the doublet is diagonalised; the comment notes that E is large at $t = T$, so H itself separates the two states (no localisation is needed). Returns (ψ_+, ψ_-, E_+). Used by `run_ramp`, `evolve_states` and `evolve_states_single_modes` to project the evolved state and to identify which logical state ended in $+E$.

7.3 Time-dependent Hamiltonian (cell 28)

`smoothstep(τ)`

```
smoothstep(τ) = τ <= 0 ? 0.0 : τ >= 1 ? 1.0 : τ^2 * (3 - 2τ)
```

$s(\tau) = 3\tau^2 - 2\tau^3$ clamped to $[0,1]$; $s'(0) = s'(1) = 0$.

`mu_xt(x, t, T, ramp; α = alpha)`

```
ℓ_center = 1                              # sites kept topological in the middle
N_left  = (Nt - ℓ_center) ÷ 2             # 15 ramped sites on the left  (x = 1 … 15)
N_right = Nt - ℓ_center - N_left          # 16 ramped sites on the right (x = 17 … 32)
N_leg   = max(N_left, N_right)            # 16
function mu_xt(x, t, T, ramp; α = alpha)
    (x <= N_left || x > Nt - N_right) || return μ_topo_t
    i = min(x - 1, Nt - x)
    q = (t/T) * (1 + α * (N_leg - 1)) - α * i
    return μ_topo_t + (μ_triv_t - μ_topo_t) * ramp(q)
end
```

Implements $\mu(x,t)$ of the protocol ($\ell_{\text{center}} = 1$, `Nt = 32`, $\alpha = 0.025$). A site is ramped if it lies in the left leg ($x \le$ `N_left`) or the right leg ($x > N_t -$ `N_right`); its delay index is the distance from the nearer chain end, $i = \min(x-1, N_t - x)$, so the two walls move inward at the same rate. With $\ell_{\text{center}} = 1$ the 31 ramped sites split into 15 (left, $i = 0..14$) and 16 (right, $i = 0..15$), and $N_{\text{leg}} = \max(15,16) = 16$ makes the innermost site of the longer leg finish exactly at $t = T$; site 16 is never ramped. The same function works for even ℓ_{center} (equal legs). The argument `ramp` is the profile function (`smoothstep`).

`onsite_mpo(V)`

```
function onsite_mpo(V)
    arr = zeros(ntuple(_ -> 2, Lt)...)
    for I in CartesianIndices(arr)
        b = Tuple(I) .- 1   # position bits, most significant first
        arr[I] = V(1 + sum(b[k] * 2^(Lt - k) for k in 1:Lt))
    end
    D = TensorBinding.mps_to_diagonal_mpo(MPS(arr, pos_t; cutoff=1e-16), pos_t)
    return TensorBinding.prepend_nambu(D, nambu_t, :tz)
end
```

Builds the exact on-site MPO $\tau_z \otimes \operatorname{diag} V(x)$ from the N_t values $V(x)$. The values are written into a dense $2\times\cdots\times2$ array indexed by the position bits, compressed to an MPS on the position sites (`pos_t = st[2:end]`) with cutoff 10^{-16}, turned into a diagonal MPO, and given the Nambu structure τ_z on `nambu_t = st[1]`. Per the comment, this avoids rebuilding the whole Hamiltonian (and running QTCI) at every time step.

`H_of_t_alt(t, T, ramp), pbt(x)`

```
H_of_t_alt(t, T, ramp) = +(H_static.mpo, onsite_mpo(x -> mu_xt(x, t, T, ramp)); cutoff=1e-16)
pbt(x) = basis_state(st, vcat([0], to_bits(x - 1, Lt)))
```

$H(t) = H_{\text{static}} + \tau_z \otimes \operatorname{diag} \mu(x,t)$, where `H_static` (hopping and pairing, no on-site term) is built once with `get_Hamiltonian("chain_1d", 1.0; L = Lt)` and `add_superconductivity!(H_static, 1.0)`; `st = H_static.sites`. `pbt(x)` is the particle-sector basis state at x. Cell 28 checks $\max_x |\langle 0,x|H|0,x\rangle - \mu(x,t)|$ for $T = 80$:

t	max on-site error	`maxlinkdim(H)`
0	1.8×10^{-15}	5
40	3.4×10^{-14}	8
80	1.1×10^{-14}	6

`norm_Hψ(H, ψ)`

```
norm_Hψ(H, ψ) = norm(orthogonalize(apply(H, ψ; alg="naive", truncate=false), 1)[1])
```

$\|H\psi\|$ computed without truncation: $H\psi$ is formed exactly, brought to orthogonality centre 1, and the norm of the first tensor is the norm of the MPS. Used once to check $\psi_f(0)$.

7.4 Evolution of the fermion mode

`run_ramp(T, ramp; dt=0.1, nsnap=100)`

```
nsteps = round(Int, T / dt); dt = T / nsteps
states = TensorBinding.evolve_with_tdvp_timedep(t -> H_of_t_alt(t, T, ramp), ψf_init, nsteps,
 ↪  dt;
                                                maxdim=64, cutoff=1e-20, tol=1e-12, krylovdim=30,
                                                   reverse_step=true)
ψ₊, ψ₋, E₊ = final_branches(H_of_t_alt(T, T, ramp), st)
ψT = states[end]
p₊, p₋ = abs2(inner(ψ₊, ψT)), abs2(inner(ψ₋, ψT))
σ = p₋ > p₊ ? -1 : 1
σ == -1 && ((p₊, p₋) = (p₋, p₊))
idx = unique(round.(Int, range(1, nsteps + 1; length=nsnap + 1)))
return (T = T, leak = 1 - p₊ - p₋, p_wrong = p₋, E₊ = E₊,
        ΔQ_final = σ * ΔQ_mode(ψT, st), ΔQ_adiabatic = ΔQ_mode(ψ₊, st),
        times = (idx .- 1) .* dt, ΔQ_t = [σ * ΔQ_mode(states[i], st) for i in idx], snapshots =
         ↪  states[idx])
```

Computes the evolution $i\partial_t|\psi(t)\rangle = H(t)|\psi(t)\rangle$ of $\psi_f(0)$ from 0 to T and the resulting non-adiabatic error and $\Delta Q(t)$.

Time stepping. `dt` is adjusted so that `nsteps*dt = T`. From the package source, the routine `evolve_with_tdvp_timedep` evaluates H at the step midpoint $t_{\text{mid}} = (k - \frac{1}{2})\,dt$, uses two-site TDVP by default (`nsite = 2`), normalises after each step by default, and returns all `nsteps+1` states. The notebook sets `maxdim = 64`, `cutoff = 1e-20`, Krylov tolerance 10^{-12} with Krylov dimension 30, and `reverse_step = true` (the comment: a sweep consists of a forward and a backward pass through the MPS).

Analysis.

- $p_\pm = |\langle\psi_\pm|\psi(T)\rangle|^2$ with the instantaneous branches from `final_branches`.
- Branch identification: if the state ended mainly in ψ_-, then $\sigma = -1$ and $p_\pm$ are swapped. Since $\Delta Q(C\psi) = -\Delta Q(\psi)$, multiplying by σ reports ΔQ for the logical state that ends in $+E$ ($|1\rangle$).
- `leak` $= 1 - p_+ - p_-$: the probability transferred into bulk quasiparticles. `p_wrong` $= p_-$ after the swap: the weight in the other branch.
- `ΔQ_final` $= \sigma\Delta Q(\psi(T))$; `ΔQ_adiabatic` $= \Delta Q(\psi_+)$, the adiabatic value.
- `times`, `ΔQ_t`, `snapshots`: `nsnap+1` (after `unique`) roughly equally spaced samples.

Used in cell 31 for $T = 150, 200, 250, 300, 350$, which prints `leak`, `p_wrong`, `ΔQ_final`, `ΔQ_adiabatic` and their difference (results in §8), and in cell 32 (Figure 5).

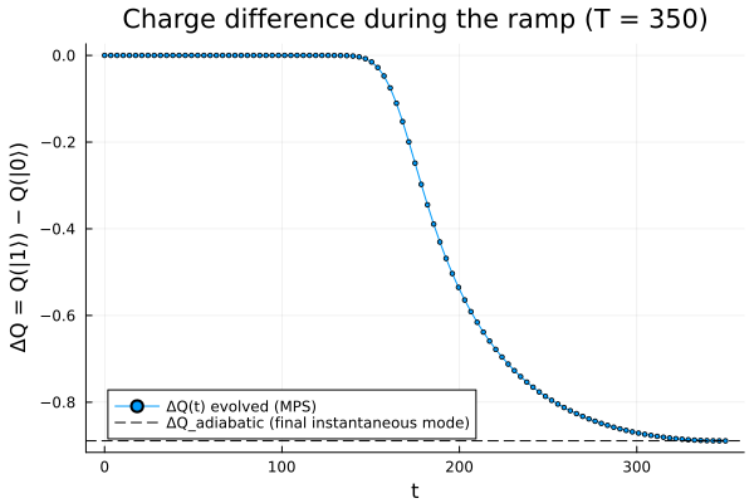


Figure 5: $\Delta Q(t)$ of the evolved ψ_f for the smoothstep ramp with $T = 350$ (`run_ramp`). ΔQ stays at zero while the MZMs are far apart and goes to -0.889 as they fuse at the centre. Dashed: the adiabatic value `ΔQ_adiabatic` $= \Delta Q(\psi_+) = -0.889108$ of the final instantaneous mode.

`evolve_states(T, ramp; dt=0.1, nsnap=160)`

Computes $\delta n(x,t) = n_1(x,t) - n_0(x,t) = |u_f(x,t)|^2 - |u_{\bar{f}}(x,t)|^2$ for both logical states.

```
s1, s0 = ev(ψf_init), ev(C_t(ψf_init))
ψ₊, _, _ = final_branches(H_of_t_alt(T, T, ramp), st)
abs2(inner(ψ₊, s0[end])) > abs2(inner(ψ₊, s1[end])) && ((s1, s0) = (s0, s1))   # |1⟩: mode
↪  ends in +E
upart(ψ) = [abs2(inner(pbt(x), ψ)) for x in 1:Nt]
δn = reduce(hcat, [upart(s1[i]) .- upart(s0[i]) for i in idx])
return (T = T, times = (idx .- 1) .* dt, δn = δn,
        C_err = norm(add2(C_t(s1[end]), -1 * s0[end])),
        n_mode = (abs2(inner(ψ₊, s1[end])), abs2(inner(ψ₊, s0[end]))))
```

Method. Both ψ_f and $\bar{\psi}_f = C\psi_f$ are evolved by TDVP with the settings of `run_ramp`. The one whose final state has more weight on ψ_+ is labelled $|1\rangle$. Per the comments: the bulk cancels exactly in $\Gamma_1 - \Gamma_0$, and evolving and subtracting the two full Γ MPOs would bury δn (about 10^{-14} early on) under truncation error; $\bar{\psi}_f$ is evolved independently rather than taken as $C\psi_f(t)$, so that $\|C\psi_f(T) - \bar{\psi}_f(T)\|$ (`C_err`) checks the evolution. `n_mode` holds the weights of the two final states on ψ_+.

Returns `(T, times, δn, C_err, n_mode)`, with `δn` an $N_t \times n_{\text{snap}}$ matrix. **Used** in cell 33 for $T = 300$ and $T = 100$ (Figure 6); the cell prints `C_err` and `n_mode` for both runs (§8).

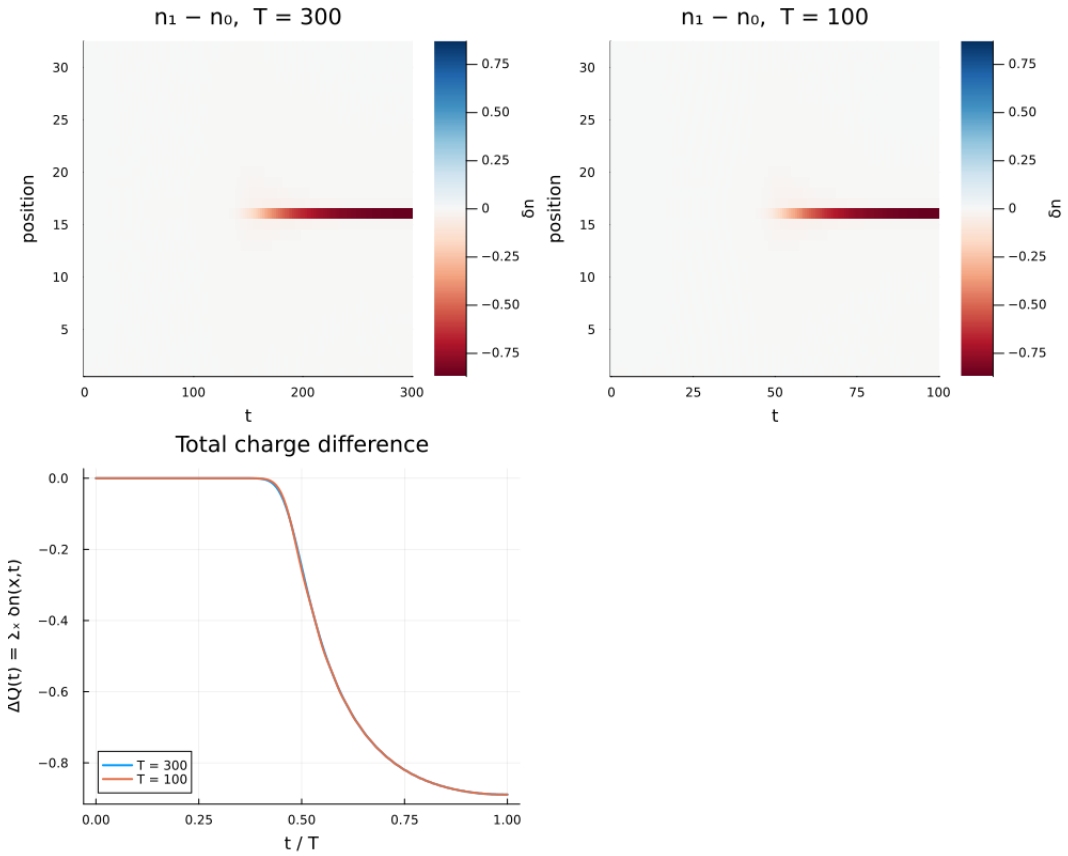


Figure 6: Top: $\delta n(x,t) = n_1 - n_0$ for $T = 300$ and $T = 100$ (`evolve_states`). The density difference appears at the centre site $x = 16$ once the MZMs come together. Bottom: $\Delta Q(t) = \sum_x \delta n(x,t)$ versus t/T; the two ramp durations give almost the same curve, ending near -0.89.

7.5 Bulk states and full densities (cells 35–39)

`bdg_basis, dense_bdg(Hmpo), hbt(x)`

```
bdg_basis = [basis_state(st, vcat([nb], to_bits(x - 1, Lt))) for nb in 0:1 for x in 1:Nt]   #
 ↪  index nb·Nt + x
dense_bdg(Hmpo) = [inner(a', Hmpo, b) for a in bdg_basis, b in bdg_basis]
hbt(x) = basis_state(st, vcat([1], to_bits(x - 1, Lt)))
```

`bdg_basis` lists the $2N_t = 64$ BdG basis states, ordered as $(u_1, \ldots, u_{N_t}, v_1, \ldots, v_{N_t})$, i.e. index $n_b N_t + x$. `dense_bdg` converts an MPO to the dense $2N_t \times 2N_t$ BdG matrix by matrix elements. `hbt(x)` is the hole-sector basis state, used to read $|v(x)|^2$.

Convention check. The cell also builds `Hs_d = dense_bdg(H_static.mpo)` and compares it with the reference matrix $[h\ \Delta;\ \Delta^\dagger\ -h^T]$ of §1.1 (`h_ref`, `Δ_ref`); it prints $h_{12} = 1$, $\Delta_{12} = 1$, $\Delta_{21} = -1$, $\Delta_{11} = -1.9 \times 10^{-17}$ and a maximum deviation of 3.6×10^{-15}.

`vec_to_mps(c; cutoff=1e-16)`

Inverse of the above: writes the dense BdG vector c (ordering $n_b N_t + x$) into a 2^{L_t+1} complex array indexed by [Nambu bit, position bits] and compresses it into an MPS on `st` with cutoff 10^{-16}.

`initial_bulk_states(Hmpo)`

```
Hd = dense_bdg(Hmpo)
F = eigen(Hermitian((Hd + Hd') / 2))
```

```
doublet_idx = sortperm(abs.(F.values))[1:2]
bulk_idx = [n for n in eachindex(F.values) if F.values[n] < 0 && n ∉ doublet_idx]
@assert length(bulk_idx) == Nt - 1
return [vec_to_mps(F.vectors[:, n]) for n in bulk_idx]
```

The filled bulk states at $t = 0$: all negative-energy eigenstates of $H(0)$ except the near-zero doublet (the two smallest $|E|$), $N_t - 1 = 31$ of them. For $N_t = 32$ the BdG matrix is only 64×64, so the comment notes that exact diagonalisation is used and the eigenvectors are converted to MPS. Stored as `bulk_init`. Printed check: $\|P_{\text{bulk}}\psi_f\|^2 = \sum_b |\langle b|\psi_f\rangle|^2 = 1.8 \times 10^{-25}$ at $t = 0$.

`bulk_density_tdvp(T, ramp; dt=0.1, nsnap=160)`

```
Hs = [H_of_t_alt((k - 0.5) * dt, T, ramp) for k in 1:nsteps]   # H at t_mid of step k
Hoft(t) = Hs[round(Int, t / dt + 0.5)]
...
s = TensorBinding.evolve_with_tdvp_timedep(Hoft, b, nsteps, dt; maxdim=64, cutoff=1e-20,
 ↪  tol=1e-12,
                                            krylovdim=30, reverse_step=true)
n_bulk[:, j] .+= upart(s[i]);  v_bulk[:, j] .+= vpart(s[i])
...
G = [inner(a, b) for a in final, b in final]
return (n_bulk = n_bulk, v_bulk = v_bulk, orth_err = maximum(abs, G - I))
```

Computes the evolved bulk density $n_{\text{bulk}}(x,t) = \sum_b |u_b(x,t)|^2$ by evolving each of the 31 bulk states with the same $H(t)$ and TDVP settings as ψ_f. $H(t_{\text{mid}})$ for every step is built once and shared by all states; `Hoft` maps the midpoint time $(k - \frac{1}{2})dt$ requested by the library back to index k. It also accumulates the hole weight $v_{\text{bulk}}(x,t) = \sum_b |v_b(x,t)|^2$ (for the completeness check) and returns the overlap error $\max_{mn} |\langle b_m|b_n\rangle - \delta_{mn}|$ of the final states, which should stay near zero. Samples are at the same snapshot indices as `evolve_states_single_modes`.

`evolve_states_single_modes(T, ramp; dt=0.1, nsnap=160)`

Same evolution and $|1\rangle$ labelling as `evolve_states`, but returns the two particle densities separately, `n_pos` $= |u_f(x,t)|^2$ (state $|1\rangle$) and `n_neg` $= |u_{\bar{f}}(x,t)|^2$ (state $|0\rangle$), together with `times`, `C_err` and `n_mode`. Used by `evolve_full_densities`.

`ramp_fermi_level(ramp; npts=401)`

```
H_static_dense = dense_bdg(H_static.mpo)
E = map(range(0, 1; length=npts)) do τ
    μ = [mu_xt(x, τ, 1.0, ramp) for x in 1:Nt]
    sort(abs.(eigvals(Hermitian(H_static_dense + Diagonal(vcat(μ, -μ))))))   # E[1] = E[2] =
     ↪  mode, E[3] = bulk edge
end
Em, Eg = maximum(e[1] for e in E), minimum(e[3] for e in E)
Em < Eg || error("no single ϵF: max E_mode = $Em ≥ min E_gap = $Eg")
return -(Em + Eg) / 2, Em, Eg, maximum(e[end] for e in E)
```

Computes one Fermi level valid for the whole ramp. Because $\mu(x,t)$ depends only on t/T, the instantaneous spectrum is scanned once over $t/T \in [0,1]$ (401 points) by exact diagonalisation of $H(t) = H_{\text{static}} + \tau_z \otimes \operatorname{diag}\mu(x,t)$, here `Diagonal(vcat(μ, -μ))` in the `bdg_basis` ordering. ϵ_F must lie below $-E_{\text{mode}}(t)$ and above $-E_{\text{gap}}(t)$ at every t, so the worst cases are used:

$$\epsilon_F = -\frac{\max_t E_{\text{mode}}(t) + \min_t E_{\text{gap}}(t)}{2}.$$

Returns $(\epsilon_F, \max_t E_{\text{mode}}, \min_t E_{\text{gap}}, \max|E|)$. Printed: $\max_t E_{\text{mode}} = 0.34875$, $\min_t E_{\text{gap}} = 0.82787$, window $(-0.82787, -0.34875)$, $\epsilon_F = -0.58831$, $\max|E| = 7.5749$.

`mcweeny_bulk(Hmpo; ϵF=ϵF_ramp, scale=9.0, maxiters=80, maxdim=200, cutoff=1e-20, tol=1e-10)`

```
Id = MPO(st, "Id")
ρ0 = +((0.5 + ϵF / (2scale)) * Id, (-0.5 / scale) * Hmpo; cutoff=cutoff)
ρ = TensorBinding.mcweeny_purify(ρ0; maxiters=maxiters, maxdim=maxdim, cutoff=cutoff, tol=tol)
abs(real(tr(ρ)) - (Nt - 1)) > 1e-6 && @warn "Tr P_bulk = $(real(tr(ρ))), expected $(Nt - 1):
 ↪  check ϵF"
return [real(inner(pbt(x)', ρ, pbt(x))) for x in 1:Nt]
```

Computes the instantaneous (adiabatic) bulk density $\langle 0, x|P_{\text{bulk}}(t)|0, x\rangle$ for a given $H(t)$. The initial guess

$$\rho_0 = \tfrac{1}{2} - \frac{H - \epsilon_F}{2\,\text{scale}}$$

has eigenvalues in $[0, 1]$ when scale $\geq \max|E - \epsilon_F|$; the comment cites $\max|E| \approx 7.6$, and scale $= 9$ is used. McWeeny $\rho \to 3\rho^2 - 2\rho^3$ drives eigenvalues above $\frac{1}{2}$ (i.e. $E < \epsilon_F$) to 1 and the rest to 0, which gives $P_{\text{bulk}}(t)$ when ϵ_F lies between the bulk edge and the lower member of the doublet. A warning is issued if $\operatorname{Tr}\rho$ differs from $N_t - 1$ by more than 10^{-6}.

`bulk_density_mcweeny(T, ramp, times; kw...)`

```
bulk_density_mcweeny(T, ramp, times; kw...) = reduce(hcat, [mcweeny_bulk(H_of_t_alt(t, T,
 ↪  ramp); kw...) for t in times])
```

Applies `mcweeny_bulk` at each snapshot time and stacks the results into an $N_t \times n_{\text{snap}}$ matrix.

`evolve_full_densities(T, ramp; bulk=:tdvp, dt=0.1, nsnap=160)`

```
m = evolve_states_single_modes(T, ramp; dt=dt, nsnap=nsnap)
if bulk == :tdvp
    b = bulk_density_tdvp(T, ramp; dt=dt, nsnap=nsnap)
    compl_err = maximum(abs, b.n_bulk .+ b.v_bulk .+ m.n_pos .+ m.n_neg .- 1)
elseif bulk == :mcweeny
    n_bulk, orth_err, compl_err = bulk_density_mcweeny(T, ramp, m.times), NaN, NaN
...
return (T = T, times = m.times, n1 = n_bulk .+ m.n_pos, n0 = n_bulk .+ m.n_neg, n_bulk =
 ↪  n_bulk,
        orth_err = orth_err, compl_err = compl_err, C_err = m.C_err, n_mode = m.n_mode)
```

Computes the full densities $n_1(x,t) = n_{\text{bulk}} + |u_f|^2$ and $n_0(x,t) = n_{\text{bulk}} + |u_{\bar{f}}|^2$, with the shared bulk either TDVP-evolved (`:tdvp`) or adiabatic (`:mcweeny`). For `:tdvp` the completeness error checks $\Gamma_1 + C\Gamma_1 C = \mathbb{1}$, i.e. at every site

$$\sum_{\text{bulk}} (|u_b(x)|^2 + |v_b(x)|^2) + |u_f(x)|^2 + |v_f(x)|^2 = 1,$$

using $|v_f|^2 = |u_{\bar{f}}|^2$. Used in cell 39 with $T = 100$ and `nsnap = 40` (Figure 7); the cell prints `orth_err`, `compl_err`, `C_err`, `n_mode` and the TDVP–McWeeny bulk difference.

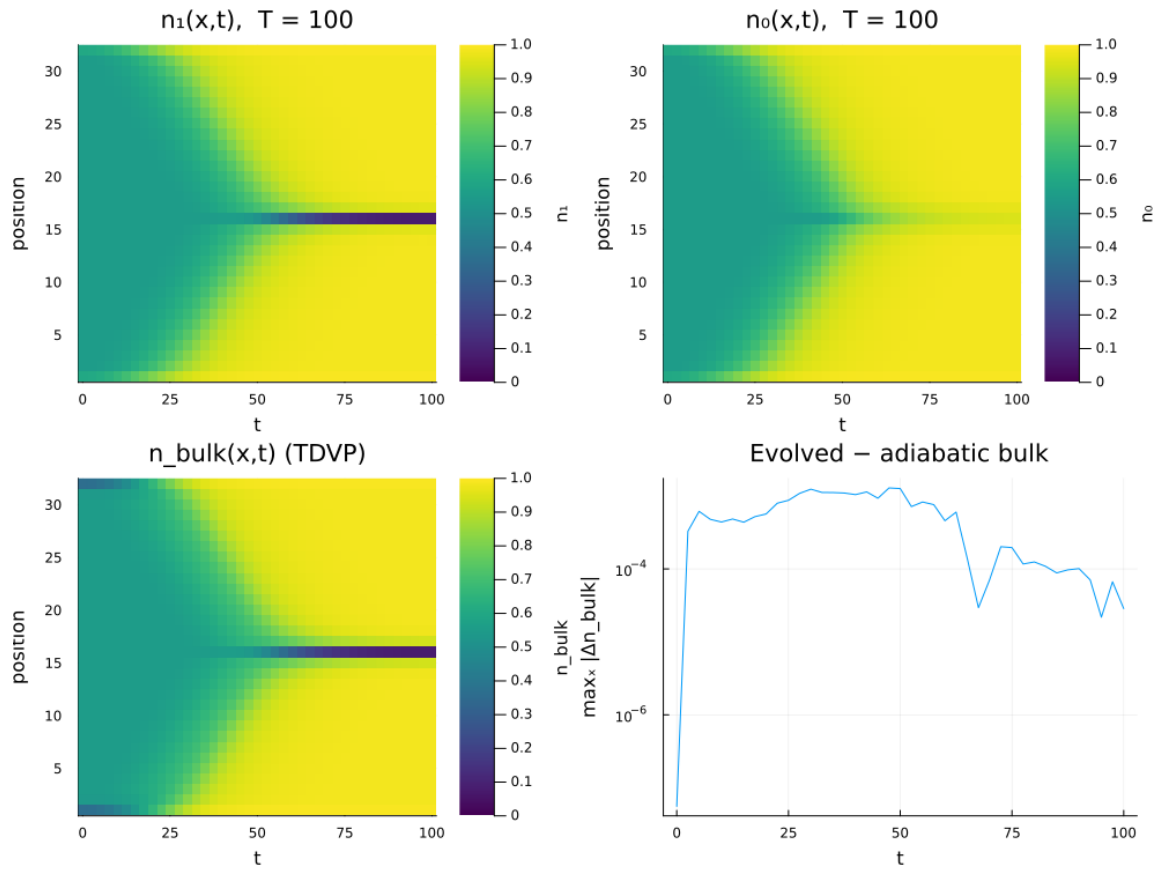


Figure 7: Full densities for the smoothstep ramp with $T = 100$ (`evolve_full_densities`, TDVP bulk). Top: $n_1(x,t)$ and $n_0(x,t)$. Bottom left: $n_{\text{bulk}}(x,t)$ from TDVP. The trivial region (density near 1) grows inward from both ends. Bottom right: $\max_x |n^{\text{TDVP}}_{\text{bulk}} - n^{\text{McWeeny}}_{\text{bulk}}|$ versus t (log scale), the difference between the evolved and the adiabatic bulk.

8 Results

8.1 Static chain, $\ell = 6$ (`H0`)

Quantity	Value
E_{mode} (`fermion_mode`)	0.0187632081
E_{gap} (`kpm_gap`)	1.1536543281
$\epsilon_F = -(E_{\text{mode}} + E_{\text{gap}})/2$	−0.5862087681
$\operatorname{Tr} P_{\text{bulk}}$	62.999999999993
$\langle\psi\|\Gamma_0\|\psi\rangle$, $\langle\psi\|\Gamma_1\|\psi\rangle$	9.0×10^{-13}, 1.0000000000009
$\langle\bar\psi\|\Gamma_0\|\bar\psi\rangle$, $\langle\bar\psi\|\Gamma_1\|\bar\psi\rangle$	0.99999999999997, -1.7×10^{-14}
$E_1 - E_0 = \frac{1}{2}\operatorname{Tr}[H(\Gamma_1 - \Gamma_0)]$	0.0187632081

8.2 ΔQ versus ℓ (`charge_difference`, cell 20)

The cell prints all 40 entries. ΔQ is negative for $|1\rangle$ (the $+E$ member). E halves with each added site from $\ell \approx 5$ on, and $|\Delta Q|$ falls by about the same factor; $|E_{\text{bulk}}|$ decreases from 2.02 to 1.006 and stays above $2E$ in every case (no gap-check warning).

ℓ	ΔQ	E	$\lvert E_{\text{bulk}}\rvert$	ℓ	ΔQ	E	$\lvert E_{\text{bulk}}\rvert$
1	-0.8481	0.7379	2.0181	21	-1.179×10^{-5}	5.722×10^{-7}	1.0235
2	-0.6107	0.322	2.0181	22	-6.180×10^{-6}	2.861×10^{-7}	1.0214
3	-0.4209	0.1536	1.7891	23	-3.233×10^{-6}	1.431×10^{-7}	1.0196
4	-0.2766	0.07557	1.5330	24	-1.688×10^{-6}	7.153×10^{-8}	1.0179
5	-0.1738	0.03759	1.3753	25	-8.798×10^{-7}	3.576×10^{-8}	1.0165
6	-0.1052	0.01876	1.2746	26	-4.578×10^{-7}	1.788×10^{-8}	1.0152
7	-0.06191	9.377×10^{-3}	1.2077	27	-2.378×10^{-7}	8.941×10^{-9}	1.0141
8	-0.03563	4.688×10^{-3}	1.1616	28	-1.234×10^{-7}	4.470×10^{-9}	1.0131
9	-0.02016	2.344×10^{-3}	1.1288	29	-6.399×10^{-8}	2.235×10^{-9}	1.0122
10	-0.01125	1.172×10^{-3}	1.1048	30	-3.307×10^{-8}	1.118×10^{-9}	1.0114
11	-6.211×10^{-3}	5.859×10^{-4}	1.0868	31	-1.710×10^{-8}	5.588×10^{-10}	1.0107
12	-3.398×10^{-3}	2.930×10^{-4}	1.0730	32	-8.829×10^{-9}	2.794×10^{-10}	1.0100
13	-1.846×10^{-3}	1.465×10^{-4}	1.0622	33	-4.356×10^{-9}	1.397×10^{-10}	1.0094
14	-9.961×10^{-4}	7.324×10^{-5}	1.0535	34	-2.268×10^{-9}	6.985×10^{-11}	1.0088
15	-5.347×10^{-4}	3.662×10^{-5}	1.0466	35	-1.187×10^{-9}	3.492×10^{-11}	1.0084
16	-2.856×10^{-4}	1.831×10^{-5}	1.0409	36	-6.271×10^{-10}	1.746×10^{-11}	1.0079
17	-1.520×10^{-4}	9.155×10^{-6}	1.0361	37	-3.124×10^{-10}	8.731×10^{-12}	1.0075
18	-8.057×10^{-5}	4.578×10^{-6}	1.0322	38	-1.619×10^{-10}	4.366×10^{-12}	1.0071
19	-4.257×10^{-5}	2.289×10^{-6}	1.0288	39	-8.226×10^{-11}	2.183×10^{-12}	1.0067
20	-2.243×10^{-5}	1.144×10^{-6}	1.0260	40	-4.882×10^{-11}	1.091×10^{-12}	1.0064

8.3 Time-dependent ramp

Ramp-time scan (`run_ramp`, smoothstep, cell 31).

T	leak	wrong-branch weight	$\Delta Q(T)$	`ΔQ_adiabatic`	$\Delta Q(T)-$`ΔQ_adiabatic`
150	1.291×10^{-7}	9.0×10^{-18}	-0.889103	-0.889108	4.43×10^{-6}
200	5.326×10^{-8}	2.2×10^{-21}	-0.889106	-0.889108	1.98×10^{-6}
250	1.495×10^{-8}	1.2×10^{-21}	-0.889105	-0.889108	2.37×10^{-6}
300	5.493×10^{-9}	1.2×10^{-22}	-0.889103	-0.889108	4.97×10^{-6}
350	4.159×10^{-9}	3.0×10^{-22}	-0.889100	-0.889108	7.10×10^{-6}

The leak decreases as the ramp is slowed down; $\Delta Q(T)$ converges to -0.88910. The adiabatic value $\Delta Q(\psi_+) = -0.889108$ is the same for every T (the final Hamiltonian does not depend on T). The remaining difference $\Delta Q(T) - \Delta Q(\psi_+)$ of 2–7×10^{-6} does not decrease with T while the leak falls by a factor of 30, so it is not a non-adiabatic effect; it is consistent with the accumulated time-step error of the midpoint-rule TDVP ($dt = 0.1$, more steps for larger T).

Checks and full densities.

- Initial mode: $\|H(0)\psi_f\| = 1.12\times10^{-12}$, $\Delta Q(0) = 4.6\times10^{-15}$, $\|P_{\text{bulk}}\psi_f\|^2 = 1.8\times10^{-25}$.
- Fermi level for the ramp: $\epsilon_F = -0.58831$ in the window $(-0.82787, -0.34875)$; $\max|E| = 7.5749$.
- $T = 100$, TDVP bulk: overlap error 9.9×10^{-9}, completeness error 4.3×10^{-9}.
- $\max|n_{\text{bulk}}^{\text{TDVP}} - n_{\text{bulk}}^{\text{McWeeny}}| = 1.29\times10^{-3}$.
- Zero modes (cell 33): $T = 300$: $\|C\psi_f(T) - \bar{\psi}_f(T)\| = 9.5\times10^{-14}$, $|\langle\psi_+|\psi_1(T)\rangle|^2 = 0.9999999945$, $|\langle\psi_+|\psi_0(T)\rangle|^2 = 1.2\times10^{-22}$; $T = 100$: 4.5×10^{-14}, 0.9999942664, 3.7×10^{-13}. The same $T = 100$ values are printed in cell 39. ψ_f and $\bar{\psi}_f$ evolved separately stay charge-conjugate partners to 10^{-13}, and the missing weight $1 - |\langle\psi_+|\psi_1(T)\rangle|^2$ (5.7×10^{-6} for $T = 100$, 5.5×10^{-9} for $T = 300$) is the non-adiabatic leak.
- Method 1 vs method 2 over the ℓ sweep (cell 24): $\max_\ell|\Delta Q^{(1)} - \Delta Q^{(2)}| = 5.6\times10^{-16}$.